



Activated alumina sludge as partial substitute for fine aggregates in brick making

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HIGHLIGHTS

- Gainful use of toxic activated alumina sludge in making bricks.
- Up to 10% of brick clay by weight could be successfully substituted with the sludge.
- No hazardous constituents released from sludge-clay bricks in TCLP leachate.
- Brick compressive strengths decreased with sludge addition but within limits.
- Resulting bricks are recommended for low-end structural applications.

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ABSTRACT

Community water defluoridation plants primarily use activated alumina (AA) for treatment. However, this process generates toxic AA-sludge for which no sustainable method of disposal is currently known. This study evaluated whether AA-sludge could be partially used in manufacturing bricks that are structurally stable and non-polluting. The AA-sludge was variably mixed with soil (0%, 5% and 10% by weight) and fired in a field kiln to produce bricks. These bricks were extensively tested for their structural, non-structural and leaching properties. Up to 10% replacement of conventional clay soil with AA-sludge could be recommended for low-end structural applications.

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1. Introduction

Sludge disposal from water treatment plants (WTPs) is a critical problem faced worldwide. The physico-chemical properties and quantities of such sludge are governed by the quality of raw water received as well as the methods employed in water treatment [1]. Typical sludge contents are inorganic in nature [2,3] but may also contain organic carbon (C) and inorganic nitrogen (N) at 3% and 0.6% by weight, respectively, similar to the C and N contents in soil [3]. The WTP sludge is primarily landfilled due to its inert and non-hazardous nature, as is largely practised in the developed countries

[4,5]. However, when groundwater containing geogenic contaminants like fluoride (F), arsenic, iron, nitrate and salinity are treated in WTPs the sludge generated is often hazardous in nature and is usually not recommended for landfilling [6].

Geogenic contamination of groundwater is an alarming problem in several regions of the world. In India, this problem is compounded as nearly 85% of the population depend on groundwater for their potable needs [7]; an estimated 12.1 million people are directly affected by poor groundwater quality [8]. Specifically, F contamination impacts nearly 20 out of 29 Indian states, with the state of Rajasthan having nearly all districts as severely impacted [9]. High fluoride (F) contamination of groundwater also affects other parts of the world like Kenya [10], Tanzania [11], Ghana [12], China [13], and Argentina [14]. Several F-removal technologies based on coagulation-precipitation, adsorption, membrane-separation and ion-exchange processes have been developed [15]. Of these technologies, activated alumina

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